Controls

1 Operating condition

Constant rated load (1)

2 Oil samples to examine

Mixed: healthy + every fault type … (2)

(3) Run simulation + diagnose

Or examine one record

Load measured DGA file

Any record

(4) Pick random record

Record number (row)

10

Diagnose selected record

Gas	ppm
H2	78
CH4	20
C2H6	11
C2H4	13
C2H2	28

Restart ML service

Showing sampling event 3: record IECTC10_0010.

FAULT INDICATED - D1: discharges of low energy (5)

3 of 4 methods name D1 (a clear majority). 1 of 5 gases are above their typical values. It is a electrical fault type (IEC 60599 Table 1).

What to do: Discharges of low energy: look at the trend closely. Confirm with a repeat sample and the gas trend before acting; if the gases rise, plan an inspection.

(13) **Agreement: methods disagree (2 of 3 agree with ML)**

(6)

Rogers ratio (IEEE C57.104)	IEC 60599 ratio method	Duval Triangle 1	Machine learning (Random Forest)	Known label of the record (7)
ND	**D1**	**D1**	**D1**	**D1**
no diagnosis: the ratios fit no coded fault Known label: wrong	discharges of low energy Known label: right	discharges of low energy Known label: right	discharges of low energy Known label: right	discharges of low energy
NO ANSWER	AGREES	AGREES	AGREES	from the data file

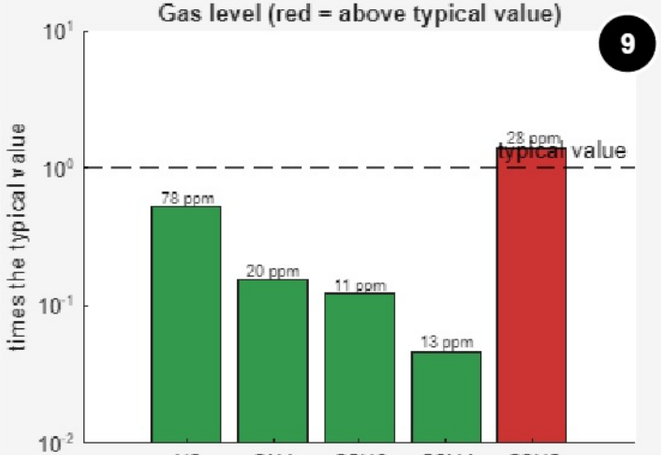
(8)

#	Time h	Record	Known label	Rogers	IEC 60599	Duval	ML	Agree of 3	Verdict	Hot-spot degC
1	3.0000	DISTEST_0065	Normal	Normal	Normal	Normal	Normal	3	HEALTHY	96.9190
2	6.0000	LI2019_0007	PD	PD	PD	PD	PD	3	FAULT PD	99.9712
3	9.0000	IECTC10_0010	D1	ND	D1	D1	D1	2	FAULT D1	100.2482
4	12.0000	IECTC10_0069	D2	D2	D1/D2	D2	D1	1	FAULT D2	100.2733
5	15.0000	DISTEST_0035	T1	ND	T1	T2	T1	1	FAULT T1	100.2756
6	18.0000	DISTEST_0042	T2	T3	T2	T3	T2	1	CHECK	100.2758
7	21.0000	IECTC10_0112	T3	T3	T3	T3	T3	3	FAULT T3	100.…

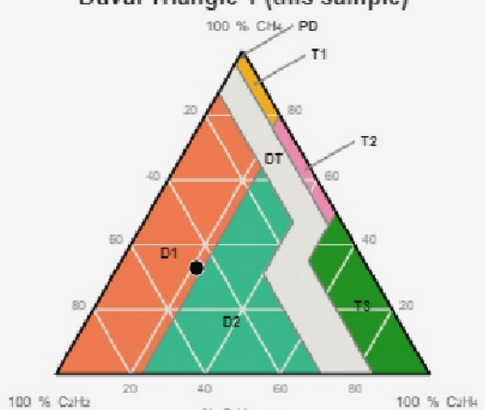
(12)

Charts | Step by step | Explanation in words | Tables | Simulation (operating condition)

Gas level (red = above typical value) (9)



(10) Duval Triangle 1 (this sample)



(11) ML probability of each fault type

